



Virtual Crowds: An LSTM-Based Framework for Crowd Simulation

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Abstract. Social modeling of pedestrian dynamics is a key element to understand the behavior of crowded scenes. Existing crowd models like the Social Force Model and the Reciprocal Velocity Obstacle, traditionally rely on empirically-defined functions to characterize the dynamics of a crowd. On the other hand, frameworks based on deep learning, like the Social LSTM and the Social GAN, have proven their ability to predict pedestrians trajectories without requiring a predefined mathematical model. In this paper we propose a new paradigm for crowd simulation based on a pool of LSTM networks. Each pedestrian is able to move independently and interact with the surrounding environment, given a starting point and a destination goal.

Keywords: Crowd modeling · Crowd simulation · LSTM · Behavior analysis

1 Introduction

Although a considerable amount of research has been carried out in the domain of behavior understanding and crowd motion analysis, there is still lack of a unified framework for validation. This is due to the fragmentation and heterogeneity of datasets used for testing and benchmarking, which often suffer from scarcity of training and testing data. Furthermore, quality of videos, size of datasets, duration of sequences, content, density of the crowd, and quality of the annotation, are only a few of the multiple factors that make it difficult to critically evaluate the performance of machine vision algorithms. To address this problem, one of the possibilities is to rely on simulators, which, although they might not resemble the scenes as natural as the real videos, they provide a considerable number of advantages, as automatic data annotation, and full control of camera parameters. The integration of simulators in the processing pipeline has been little investigated in the existing literature, and has initially been ruled by purely agent-based models [5], thus conducting the analysis on the agents ground-truth position.

The use of simulators in computer vision is not new per se, in particular for object tracking and camera control [16], and, in computer vision, we have recently observed an increasing interest in the generation of virtual videos instead

of virtual agents behaviors. The substantial difference between the two consists of carrying out the analysis with standard computer vision techniques, using the videos generated by the simulator for testing purposes. This allows taking into account potentially all the challenges of a real video, including the presence of occlusions, obstacles, changes in the illumination conditions, similarity of objects in the appearance model, etc.

In pedestrian dynamics, researchers have focused in the past on the study of trajectories and social interactions between agents. The Social Force Model (SFM) [10], the Reciprocal Velocity Obstacle model [20], and the continuum crowds model [19] are among the best existing empirical models for crowd simulation derived from observation of the real world.

Recently, deep learning has been applied to prediction and forecasting of agents in a video. Recurrent Neural Networks (RNNs), and in particular Long-Short Term Memory (LSTM) networks, have successfully been adopted to predict the scene evolution over time [1]. This kind of networks has shown the ability to learn the relation between spatially distributed data and its evolution. LSTMs has also proven to be capable of generating video sequences complying with predefined patterns [7]. Starting from the work in [1], we extend the use of LSTM networks from path prediction to simulation of crowded video sequences. The proposed method is implemented via a recurrent deep neural network based on LSTM cells. Each agent in the simulation is driven by its own LSTM network, which is aware of the hidden state of neighbouring agents. We train the LSTM on our synthetic dataset, showing how the network is effectively able to learn and replicate the simulated motion model. This demonstrates that gathering person-specific datasets of real subjects would open to the opportunity to learn person-specific behaviors, allowing for the inclusion of personal inclinations in the virtual crowd model for a more realistic simulation.

To prove the feasibility of our approach in simulating end-to-end trajectories, we validate our model using a synthetic dataset where agents move according to the SFM [10], demonstrating how the proposed framework can effectively learn the typical features of the SFM and reproduce them on different agents. The compliance of the predicted paths against the SFM is measured using the spatial distribution metric introduced in [9].

2 Related Work

The so-called Social Force Model relies on Newtonian forces to model the interaction between pedestrians and to guide each agent towards its goal. Similarly, in robotics, the Reciprocal velocity Obstacle (RVO) [20] is widely used to model the collision avoidance behavior between agents.

In the work by Treuille et al. [19], the crowd is modeled as a fluid, where agents are influenced by their goal, position, their preferred speed and a discomfort factor. However, the models mentioned above require the estimation or computation of parameters, in order to replicate the interaction among agents. The literature has tackled this issue relying on common optimization methods,

like the Gradient-based Newton method [17] and Genetic Algorithms [15]. All these methods are able to describe crowd behaviors assuming the availability of a strong prior for the model. While they are good in describing the movements of the whole crowd, they often fail in correctly representing the agent’s personality, which is a key feature to enrich the model and make the simulation closer to the behavior of real subjects. Data-driven approaches have recently been employed to capture and model interactions among people in a crowd. Patil et al. [14] use vector fields, either learned or manually sketched, to guide the crowd flow in the environment. Goal-dependent velocity fields have also been used to guide the simulation at a global level [3]. Lerner et al. [12] guide the agents using learned trajectories, choosing the one that best matches the situation that the agent is facing. Empirically-defined models can only tweak the behavior of different agents by varying parameters such as personal distances and preferred velocities; data-driven simulations allow instead learning and simulating common social rules, such as the behavior at a roundabout, but local descriptors for each agent personality cannot be handled directly. Cognitive models have been employed to allow agents performing high-levels tasks such as path planning [6]. Various approaches have been employed to vary the pedestrian behavior in the same situation, such as the OCEAN model [5] and the general adaptation syndrome [2]. These models focus on local shaping of the motion, based on personality traits, rather than long-term tasks.

As far as path prediction and activity forecasting is concerned, the literature reports a good amount of relevant works. The Social Force Model has shown good results in predicting pedestrian trajectories in the environment. Pellegrini et al. have proposed the Linear Trajectory Avoidance model [15], which predicts the short term path of a pedestrian. Interactive Gaussian processes have shown the ability to effectively improve the capability of a robot to predict pedestrian trajectories and navigate a crowded environment [18]. More recently, Recurrent Neural Networks, in particular LSTMs, have proven good capabilities in describing spatially distributed data, also providing a temporal link in sequences. LSTMs have been applied to crowd trajectory prediction by Alahi et al. [1]. However, as mentioned above, one important issue in crowd simulation is the lack of a recognized evaluation metric to assess the quality of a simulation. If the simulation refers to an existing, real-life scenario, it can be evaluated using common tools, by measuring global features, such as density and distribution of the elements in the scene. Density-based [13] and entropy-based metrics [8] are often used to compare real and synthetic data.

In [9], Helbing et al. present a metric based on the density distribution between two sets of trajectories. As detailed later, we will adopt this metric to evaluate the performances of our simulator in a quantitative manner.

3 The Proposed Model

Empirically-defined parametric functions are generally good in reproducing the global motion properties of the crowd, but they tend to fail when capturing the

complexity that leads each human to react differently when facing the presence of other subjects or obstacles.

In the domain of path prediction, LSTM networks [11] have shown good capabilities in predicting the behavior of pedestrians, if properly trained. In particular, the Social LSTM model [1] is able to capture the status of the neighborhood around each agent (number and position of other agents), which is then used to refine the trajectory prediction. In our work, we propose a model able to perform a simulation of a crowded scene by learning the motion properties of the agents from a set of training video sequences. Each agent in the simulation is provided with an autonomous network, based on LSTM cells. Each pedestrian tries to reach its target destination given a starting point, a goal and the status of his neighborhood.

Depending on the set of trajectories used to train the specific network, each LSTM network can react in a different way to the same scenario. For example, if an LSTM network is trained with a set of trajectories belonging to a specific person, it will then be able to learn that model, thus replicating the subject reactions (personality traits) to specific situations occurring in the simulation. The possibility of reproducing different personality traits makes it possible to simulate semantically rich realistic situations, which do not rely on purely deterministic models. To the best of our knowledge, such paradigm has not been explored in the state-of-the-art and no dataset in this form has been made available to the research community.

3.1 Virtual Crowds

Extending the LSTM paradigm to crowd simulation, each agent's movements are controlled by an LSTM network, able to guide it through the environment.

Agents (pedestrians in our scenario) moving in the crowd are influenced by their goal, their personality, and the state of their neighborhood. All necessary details are provided in the next paragraphs.

Goal Modeling. At the beginning of the simulation each agent/pedestrian ped_i , whose position is defined as (x_t^i, y_t^i) , is initialized at a starting position (x_0^i, y_0^i) . The goal of the i -th pedestrian ped_i is defined as

$$g^i = (x_g^i, y_g^i) \quad (1)$$

where x_g^i and y_g^i are coordinates defined in meters in the simulator.

The chain structure of a cell of the neural network is shown in Fig. 1. At each time step, the inputs of the LSTM cell representing a pedestrian are (i) the previous position (x_{t-1}^i, y_{t-1}^i) , (ii) the goal g^i , and (iii) the Social pooling tensor H_t^i . The output of the LSTM cell is the agent position (x_t^i, y_t^i) .

When the Euclidean distance between the current position (x_t^i, y_t^i) and the goal g^i is less than one meter, the simulation of that pedestrian is terminated.

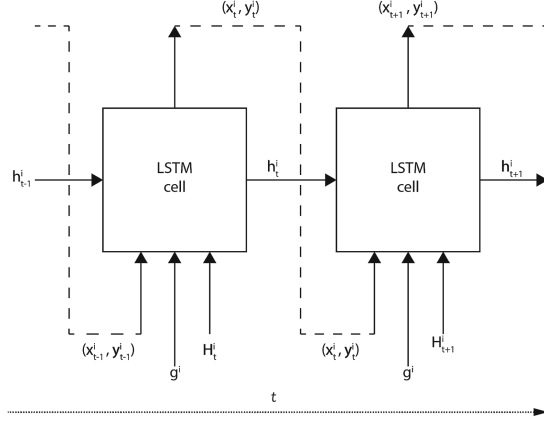


Fig. 1. The figure represents the chain structure of the LSTM network between two consecutive time steps, t and $t + 1$. At each time step, the inputs of the LSTM cell representing a pedestrian are the previous position (x_{t-1}^i, y_{t-1}^i) , the goal g^i and the Social pooling tensor H_t^i . The output of the LSTM cell is the current position (x_t^i, y_t^i) .

Neighborhood Modeling. The state of the neighborhood of each pedestrian is represented by a “Social” hidden-state tensor, as proposed by [1]. The Social pooling layer allows pedestrians to share their hidden states, thus enabling each network to predict the next position, reasoning about its hidden state and the neighborhood state.

The pedestrian ped^j at time t in the scene is represented by the hidden-state h_t^j of a LSTM network. We choose the hidden-state dimension D and the neighborhood size N_0 . The neighborhood of the agent ped^i is handled by a “Social” pooling layer. This layer has the aim of pooling information received from the LSTM cells of its neighbors while preserving also their spatial mapping. This spatial mapping is preserved through a grid pooling as explained in Eq. 3. The pooled information is used to build a tensor, called “Social” hidden-state tensor H_t^i , with dimensions $N_0 \times N_0 \times D$:

$$H_t^i(m, n, :) = \sum_{j \in N_i} 1_{mn}[x_t^j - x_t^i, y_t^j - y_t^i] h_{t-1}^j \quad (2)$$

where h_{t-1}^j represents the hidden-state of the LSTM of ped^j ($\forall j \neq i$) at $t - 1$, N^i represents the set of neighbors of pedestrian ped^i and $1_{mn}[x, y]$ is an indicator function defined as:

$$1_{mn}[x, y] = \begin{cases} 0 & \text{if } [x, y] \notin \text{cell } mn \\ 1 & \text{if } [x, y] \in \text{cell } mn \end{cases} \quad (3)$$

A graphical representation of the pooling operation is shown in Fig. 2.

Once computed, the Social hidden-state tensor is embedded into a vector a_t^i . The output coordinates are embedded in the vector e_t^i . The resulting recurrence is then defined by the following equations:

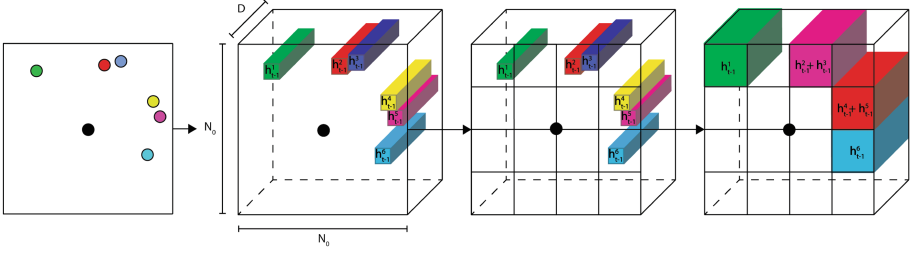


Fig. 2. Representation of the Social pooling layer. The black dot represents the pedestrian of interest ped_i . Other pedestrians ped_j ($\forall j \neq i$) are shown in different color codes. The state of the neighborhood of ped_i is described by $N_0 \times N_0$ cells, which pooling together spatially-close neighbors preserves the spatial information. This will later be used to construct the hidden-state tensor H_t^i .

$$r_t^i = \Phi(x_t^i, y_t^i; W_r) \quad (4)$$

$$e_t^i = \Phi(a_t^i, H_t^i; g^i, W_e) \quad (5)$$

$$h_t^i = LSTM(h_{t-1}^i, e_t^i; W_l) \quad (6)$$

where Φ is an embedding function with ReLU activation; W_r and W_e represent the embedding weights, and W_l represents the LSTM weights.

The next position (x_{t+1}^i, y_{t+1}^i) in the simulation depends on the hidden-state at the previous time-step h_t^i . Inspired by [7], and as performed in [1], we predict the following parameters, which characterize a bivariate Gaussian distribution: the mean $\mu_{t+1}^i = (\mu_x, \mu_y)_{t+1}^i$, the standard deviation $\sigma_{t+1}^i = (\sigma_x, \sigma_y)_{t+1}^i$ and the correlation coefficient ρ_{t+1}^i . We use a $5 \times D$ weight matrix W_p to estimate the parameters. Thus, the coordinates at the next time-step $t + 1$ are computed as:

$$(x_{t+1}^i, y_{t+1}^i) \sim N(\mu_t^i, \sigma_t^i, \rho_t^i) \quad (7)$$

In order to estimate the parameters of the LSTM model, the negative log-Likelihood loss L^i for agent ped^i is minimized for the current time instant t :

$$[\mu_t^i, \sigma_t^i, \rho_t^i] = W_p h_{t-1}^i \quad (8)$$

$$L^i(W_e, W_l, W_p) = - \sum_{t=T_{cur}+1}^{T_{step}} \log(\mathbb{P}(x_t^i, y_t^i \mid \mu_t^i, \sigma_t^i, \rho_t^i)) \quad (9)$$

Our model is trained minimizing the log-Likelihood loss for all the trajectories belonging to the dataset.

Personality. As far as the personality and decision making is concerned, the motion of the pedestrian in the crowd has been modeled in existing simulators by empirically varying parameters, such as pedestrian velocities and personal distances. Although this could actually increase the diversity of the agents' motion,

it is still not sufficient for a realistic modeling of the personality traits, which in general lead to non-deterministic and non-linear motion models. Providing a “personal” set of trajectories to each agent in the scene would allow for a much more complex and diverse simulation.

Let us consider two pedestrians in the real world, where one exhibits an *aggressive* behavior, while the other exhibits a *shy* behavior [2]. Let us assume that we are able to collect a significant set of trajectories for each of the two pedestrians. During simulation we associate ped_i as the aggressive subject, and ped_j as the shy one. In this way, if we introduce both of them in the same simulated scenario, such that the social pooling tensor is $H_t^i(m, n, :) = H_t^j(m, n, :)$, and goals are $g^i = g^j$, the two pedestrians will take different decisions, depending on the specific features.

3.2 Implementation Details

To achieve our results, we have used the following configuration. The embedding dimension for the spatial coordinates is set to 64. The spatial pooling size, which corresponds to an area of $4 \times 4 \text{ m}^2$, is set to 32. The pooling operation is performed using a sum pooling window of size 8×8 with no overlaps. The hidden state dimension is 128. The learning rate parameter is set to 0.003 and RMS-prop [4] is used as the optimizer. The model is trained on a single GPU using a PyTorch¹ implementation.

At the input of the LSTM, each trajectory consists of a set of coordinates (X_{real}, Y_{real}) in meters, which needs to be normalized. Each pair (x_{real}, y_{real}) in the set is normalized between $[-1, 1]$ using the following conversion:

$$(x_{norm}, y_{norm}) = \left(2 * \frac{x_{real} - \min(X_{real})}{Norm} - 1, 2 * \frac{y_{real} - \min(Y_{real})}{Norm} - 1 \right) \quad (10)$$

where $Norm$ represents the maximum range (in meters) of the biggest scene. This normalization is in line with the experiments conducted in previous works in this area [11].

4 Results

Dataset. To test our approach, we created a dataset composed of 5 fully annotated scenes, which visually match other known real datasets, namely the ETH [15] and UCY dataset [12], used in the experiments in [1]. An example is shown in Fig. 3. The choice of using real datasets as reference is to confront the agents’ motion to scenes known to the community, rather than reproducing all the very fine details in terms of appearance, which is in this work not relevant.

The average density of pedestrians per frame is 10. The five different scenes include 204, 177, 194, 210, and 174 trajectories, respectively. The videos have been recorded at a rate of 15 fps. The preferred velocity in the Social Forces model is set to 4.7 km/h, which corresponds to the average speed of a walking pedestrian.

¹ <http://pytorch.org>.

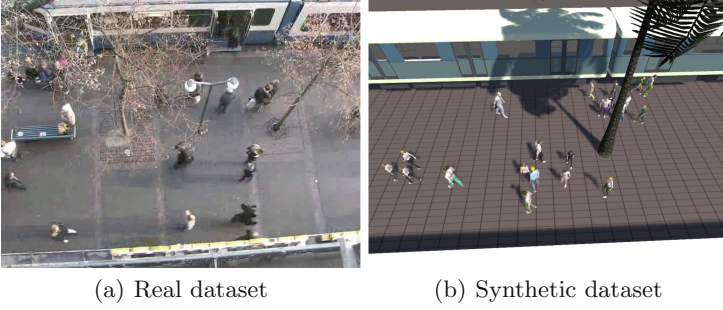


Fig. 3. A comparison of two frames of the real and synthetic datasets. The scene of interest is the so-called Hotel scene of the ETH dataset [15].

Experiments. To demonstrate the learning capabilities of the proposed framework, we rely on the simulated sequences based on SFM. Each LSTM network is trained for 100 epochs. Both training and test are conducted according to the leave-one-out strategy. This corresponds to training on 4 sets of trajectories and simulate a scene, which is similar to the fifth one. The prediction is updated at steps of 0.4s. In other words, the agent will re-consider its next displacement according to its surroundings after 6 frames.

For each agent, the simulation starts in the time instant it appears in the scene. Each pedestrian is assigned a goal, namely the last known position in the scene. Our model computes the trajectories for each pedestrian and returns the set of positions for all of them at each time step, for a complete and exhaustive representation of the scene content.

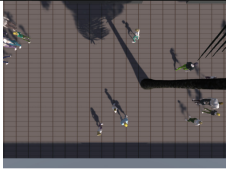
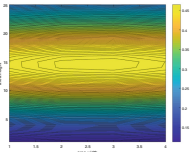
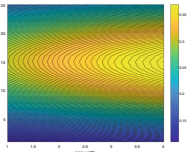
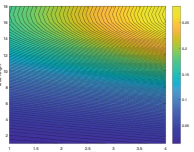
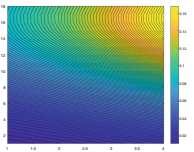
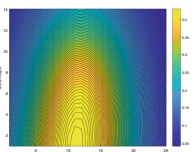
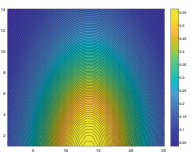
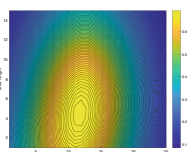
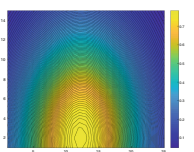
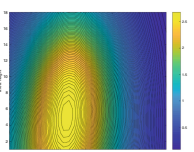
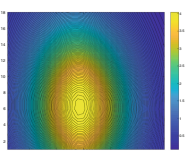
4.1 Validation

To evaluate the simulation of the model with the dataset trajectories, we compare the density distribution $\rho_{i,j}$. The density of the distribution is computed using the equation suggested by Helbing et al. [9], as shown in Eq. 11, where n represents the total number of points of all trajectories; $d(i, j, k)$ is the Euclidean distance between the k -th point along a trajectory and the (i, j) point of the grid; and R is a scaling factor:

Table 1. For each video sequence, the table reports the number of points available in the simulated video, and the number of grid cells in the horizontal (i) and vertical (j) dimensions.

	n	i	j
ETH Univ	356	4	25
ETH Hotel	383	4	18
UCY Zara1	455	25	14
UCY Zara2	654	25	15
UCY Univ	3006	30	18

Table 2. Simulation of the five different video scenes that mimic real datasets, as used in our experiments. The left column reports an overview of the scene, while the central and right columns report an example of the original video density and a sample of the one simulated using our framework. The value of ρ is color-coded: warmer colors represent higher densities, while colder colors represent lower values. Numerical values are reported on the right column at the right of each plot.

Name	Scene	Original Density	Simulated Density
ETH Hotel			
ETH Univ			
UCY Univ			
UCY Zara1			
UCY Zara2			

$$\rho_{i,j} = \frac{1}{2\pi R^2} \sum_{k=1}^n \exp\left(-\frac{d(i,j,k)^2}{R^2}\right) \quad (11)$$

The accumulation points (i, j) are displaced in a grid that covers the whole image with a distance of 1 m between them. The real dimensions of each scene, grid size (i, j) , and the number of total points n are shown in Table 1. Scaling factor is equal to 10. The plot of the density distribution of a set of trajectories for each dataset is shown in Table 2.

5 Conclusions

We have presented a framework for crowd simulation. The model assigns a neural network to each agent in the simulation. The pedestrians are connected by a social-pooling layer, which allows them to be aware of the status of their neighborhood. Compared to empirically-defined functions used in the literature to model crowds, the proposed solution allows simulating a more complex and enriched variety of behaviors. Training the network with different datasets allows in fact to reproduce the behavior of agents with different personalities. As stated at the beginning, the proposed method does not have the goal of improving the state of the art, but proposing a novel approach based on a simulator. Due to this reason, there is no comparison with the any state of the art method for trajectories, since, to the best of our knowledge, there are no other simulators in literature which we can compare to. In future works we plan to incorporate an Ego-Vision framework such that each pedestrian navigates the environment according to the information gathered from a first-person perspective. This would also allow us to better extract personality traits, since the first person view contains many more fine-grained details about the person wearing the camera.

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